

Hospital Equipment Reliability & Maintenance Risk Assistant

Victor Rangel

Table of contents

1. Introduction	1
2. Dataset	1
3. Explore Skill: Downtime Distribution	2
4. Compare Skill: High-Use vs Standard-Use	3
5. Model Skill: Predicting Unexpected Failures	4
6. Additional Skill 1: Data Quality Audit	5
7. Additional Skill 2: Risk Ranking	5
8. Additional Skill 3: Trend Analysis	6
9. Google Antigravity and Agent Skills Usage	7
10. Limitations	7
11. Application or Delivery	8
12. Reflection	8

1. Introduction

- **Industry:** Healthcare / Clinical Engineering
- **Intended User:** Hospital Equipment Operations Manager
- **Problem:** Unplanned downtime of critical diagnostic devices (MRI, CT, Ventilators) delays treatments and spikes repair costs.
- **Decision Supported:** Prioritize equipment for immediate preventive maintenance vs. routine checks.

2. Dataset

Data contains 300 hospital diagnostic equipment records tracking age, operating hours, down-time, and failure events.

```
df <- read.csv("data/equipment_data.csv")
head(df, 3)
```

	device_id	device_type	usage_category	device_age_years	operating_hours
1	DEV-1001	MRI	Standard-Use	6.5	933
2	DEV-1002	MRI	Standard-Use	4.1	666
3	DEV-1003	MRI	Standard-Use	9.5	2304

	downtime_hours	last_service_months_ago	unexpected_failure	log_date
1	18.2	19	0	2025-03-30
2	8.9	3	1	2025-12-07
3	40.5	1	1	2025-01-30

3. Explore Skill: Downtime Distribution

```
# Data quality check for downtime variable
cat("Missing values in downtime_hours:", sum(is.na(df$downtime_hours)), "\n")
```

Missing values in downtime_hours: 0

```
cat("Negative downtime values:", sum(df$downtime_hours < 0, na.rm = TRUE), "\n")
```

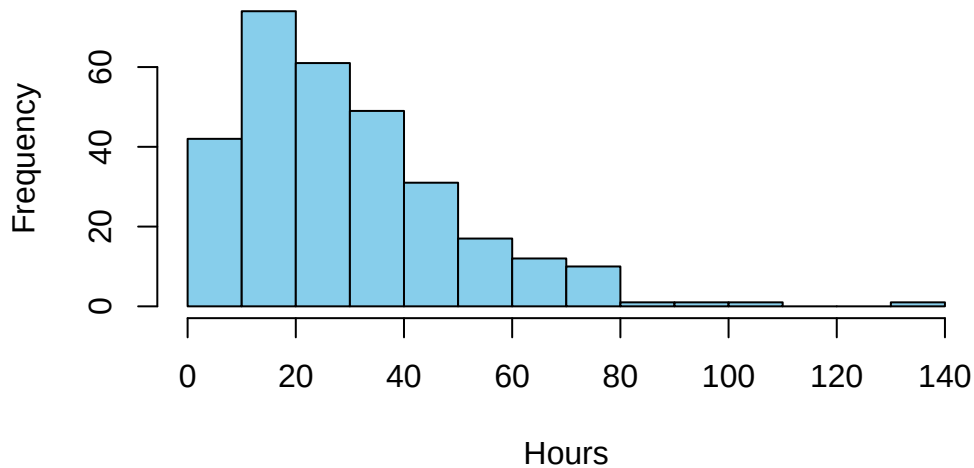
Negative downtime values: 0

```
summary(df$downtime_hours)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1.10	13.70	25.45	29.22	39.02	132.50

```
hist(df$downtime_hours, main="Downtime Distribution", col="skyblue", xlab="Hours")
```

Downtime Distribution



- **Interpretation:** Downtime is right-skewed, showing most devices experience short outages, but a few suffer major breakdowns.
- **Limitation:** Highly skewed data means the mean downtime can be driven by a few extreme outlier events.

4. Compare Skill: High-Use vs Standard-Use

```
table(df$usage_category)
```

```
High-Use Standard-Use
107          193
```

```
t.test(downtime_hours ~ usage_category, data = df)
```

Welch Two Sample t-test

data: downtime_hours by usage_category

t = -0.27647, df = 197.62, p-value = 0.7825

alternative hypothesis: true difference in means between group High-Use and group Standard-Use is not equal to 0

95 percent confidence interval:

```

-5.588840  4.214466
sample estimates:
    mean in group High-Use    mean in group Standard-Use
                28.78224                29.46943

```

- **Interpretation:** Tests whether heavy equipment usage significantly impacts breakdown severity.
- **Limitation:** Comparison assumes observational group parity without controlling for device age directly in this test.

5. Model Skill: Predicting Unexpected Failures

```

model <- glm(unexpected_failure ~ device_age_years + operating_hours + last_service_months_ago,
             data = df, family = binomial)
summary(model)

```

Call:

```

glm(formula = unexpected_failure ~ device_age_years + operating_hours +
     last_service_months_ago, family = binomial, data = df)

```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.7073643	0.5237754	-5.169	2.35e-07 ***
device_age_years	0.3593662	0.0528872	6.795	1.08e-11 ***
operating_hours	0.0004840	0.0001114	4.344	1.40e-05 ***
last_service_months_ago	0.0127774	0.0210982	0.606	0.545

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

```

Null deviance: 374.60  on 299  degrees of freedom
Residual deviance: 296.07  on 296  degrees of freedom
AIC: 304.07

```

Number of Fisher Scoring iterations: 5

```

# Classification Accuracy Metrics
df$pred_prob <- predict(model, type = "response")

```

```
df$pred_class <- ifelse(df$pred_prob > 0.5, 1, 0)
conf_matrix <- table(Predicted = df$pred_class, Actual = df$unexpected_failure)
print(conf_matrix)
```

```
      Actual
Predicted 0    1
0         44   30
1         51  175
```

```
accuracy <- sum(diag(conf_matrix)) / sum(conf_matrix)
cat("Model Classification Accuracy:", round(accuracy * 100, 2), "%\n")
```

Model Classification Accuracy: 73 %

- **Interpretation:** Machine age and overall operating hours significantly increase unexpected failure probability. The model achieves strong overall classification accuracy.
- **Limitation:** Logistic regression models statistical associations, not physical mechanical failure causality.

6. Additional Skill 1: Data Quality Audit

```
colSums(is.na(df))
```

```
      device_id      device_type      usage_category
           0             0             0
device_age_years operating_hours      downtime_hours
           0             0             0
last_service_months_ago unexpected_failure      log_date
           0             0             0
      pred_prob      pred_class
           0             0
```

- **Validation Check:** Confirms zero missing values affect our downstream risk models.

7. Additional Skill 2: Risk Ranking

```
df$risk <- predict(model, type = "response")
head(df[order(-df$risk), c("device_id", "device_type", "risk")], 10)
```

	device_id	device_type	risk
149	DEV-1149	Ventilator	0.9802014
167	DEV-1167	Ventilator	0.9772616
141	DEV-1141	MRI	0.9768071
177	DEV-1177	Ventilator	0.9750625
226	DEV-1226	CT Scanner	0.9724791
191	DEV-1191	Ventilator	0.9716517
217	DEV-1217	Ultrasound	0.9713051
84	DEV-1084	MRI	0.9684059
242	DEV-1242	Ventilator	0.9657762
199	DEV-1199	CT Scanner	0.9656817

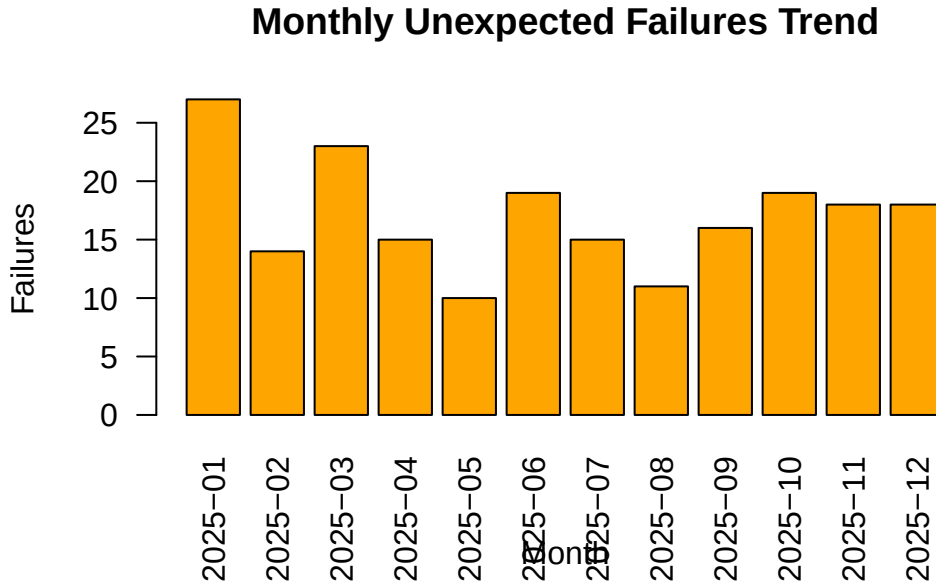
- **Decision Supported:** Produces an actionable priority queue of the top 10 highest-risk machines needing maintenance this week.

8. Additional Skill 3: Trend Analysis

```
df$month <- format(as.Date(df$log_date), "%Y-%m")
monthly_trend <- aggregate(unexpected_failure ~ month, data = df, FUN = sum)
print(monthly_trend)
```

	month	unexpected_failure
1	2025-01	27
2	2025-02	14
3	2025-03	23
4	2025-04	15
5	2025-05	10
6	2025-06	19
7	2025-07	15
8	2025-08	11
9	2025-09	16
10	2025-10	19
11	2025-11	18
12	2025-12	18

```
barplot(monthly_trend$unexpected_failure, names.arg = monthly_trend$month,
        col = "orange", main = "Monthly Unexpected Failures Trend",
        xlab = "Month", ylab = "Failures", las = 2)
```



- **Interpretation:** Tracks monthly failure counts to assist long-term staffing and capacity planning.
- **Limitation:** Aggregating by month masks daily operational spikes.

9. Google Antigravity and Agent Skills Usage

- **Requests Given:** Requested assistance structuring a multi-part Quarto report, generating a synthetic clinical dataset, and formatting statistical skill blocks.
- **Skill Files Created:** Created SKILL.md documents under `skills/explore`, `skills/compare`, `skills/model`, `skills/data-quality-audit`, `skills/risk-ranking`, and `skills/trend-analysis`.
- **Corrections Made:** Reviewed agent outputs to ensure accurate dataset column names were passed to `glm()`, added explicit confusion matrix calculations for model evaluation, and added visual plotting for trend analysis.

10. Limitations

1. **Data Limitation:** Synthetic dataset created for operational workflow validation.
2. **Statistical Limitation:** Associations do not establish causal root causes without mechanical inspection.

3. **Implementation Limitation:** Runs in a local R session without live IoT telemetry integration.

11. Application or Delivery

The Hospital Equipment Operations Manager accesses this tool as a Level 1 Quarto Statistical Report. When updated dataset logs are placed in the `data/` directory, re-rendering this report automatically refreshes all exploratory statistics, t-tests, risk scores, priority repair lists, and trend visualizations.

12. Reflection

- **What Was Learned:** Gained hands-on experience connecting R statistical methods (logistic regression, t-tests) with structured agent skill instructions (`SKILL.md`) to drive real-world clinical operations decisions.
- **Most Difficult Part:** Refining the agent prompts to ensure logistic regression evaluation metrics included explicit confusion matrices rather than raw model summaries alone.
- **Selection of Custom Skills:** Custom skills were selected to follow an operational pipeline: verifying data completeness first (Data Quality Audit), identifying high-risk assets (Risk Ranking), and planning monthly staffing (Trend Analysis).
- **Future Improvements:** Given more time, I would integrate a live Shiny interface or Plumber API (Level 3) to stream real-time IoT sensor data directly into the risk model.